



Prepared by group 1

Predicting Students' Dropout: Data - Driven Insights

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Project Context



The dataset was collected from a Portuguese higher education institution and integrates data from multiple administrative and academic information systems. It was originally compiled for a 2021 case study aimed at identifying the main factors associated with student dropout, continued enrollment, and graduation in undergraduate programs.



Correlation

The Target

Our “y” value, has 3 categories

- Dropped Out (mapped to 0)
- Currently Enrolled (mapped to 1)
- Graduated (mapped to 2)

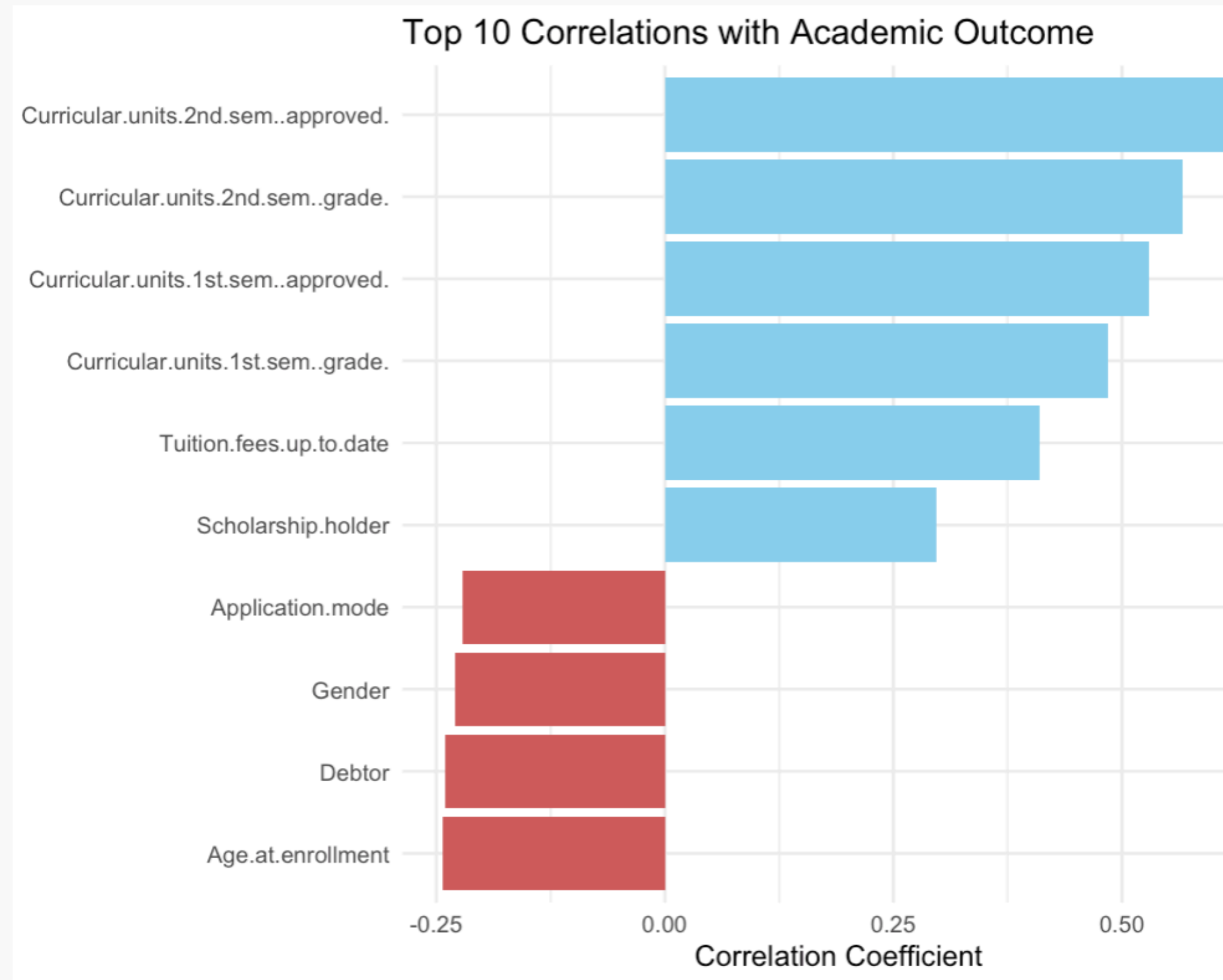
Correlation to target

- Grades and if classes passed correlate most to the target in terms of positivity
- Age and if the person owes tuition correlate most to the target for negativity



	value
Target	1.000000000
Curricular.units.2nd.sem..approved.	0.624157464
Curricular.units.2nd.sem..grade.	0.566827280
Curricular.units.1st.sem..approved.	0.529123255
Curricular.units.1st.sem..grade.	0.485207391
Tuition.fees.up.to.date	0.409826755
Scholarship.holder	0.297595291
Curricular.units.2nd.sem..enrolled.	0.175846824
Curricular.units.1st.sem..enrolled.	0.155973969
Admission.grade	0.120889157
Displaced	0.113985570
Previous.qualification..grade.	0.103763697
Curricular.units.2nd.sem..evaluations.	0.092720649
Application.order	0.089790905
Daytime.evening.attendance.	0.075106501
Curricular.units.2nd.sem..credited.	0.054003808
Curricular.units.1st.sem..credited.	0.048149715
Curricular.units.1st.sem..evaluations.	0.044361553
GDP	0.044134690
Course	0.034218826
Unemployment.rate	0.008626681
International	0.003933993
Father.s.qualification	-0.001392692
Father.s.occupation	-0.001898935
Mother.s.occupation	-0.005628565
Educational.special.needs	-0.007353073
Nacionality	-0.014801187
Inflation.rate	-0.026874065
Mother.s.qualification	-0.043177719
Previous.qualification	-0.056038592
Curricular.units.1st.sem..without.evaluations.	-0.068701819
Marital.status	-0.089803532
Curricular.units.2nd.sem..without.evaluations.	-0.094027770
Application.mode	-0.221746581
Gender	-0.229269574
Debtor	-0.240998903
Age.at.enrollment	-0.243437511

Correlation



Linear Regression

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Residuals:
    Min       1Q   Median       3Q      Max
-2.0267 -0.2840  0.1150  0.3556  2.1216

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.032e-01   1.905e-01   2.641 0.008317 **
Marital.status  4.814e-03   2.479e-02   0.194 0.846010
Application.mode -7.997e-04   9.489e-04  -0.843 0.399424
Application.order -5.433e-03   1.031e-02  -0.527 0.598176
Course          9.051e-06   8.902e-06   1.017 0.309416
Daytime.evening.attendance. -1.915e-03   4.771e-02  -0.040 0.967997
Previous.qualification  3.360e-03   1.402e-03   2.397 0.016610 *
Previous.qualification..grade.  1.676e-03   1.151e-03   1.456 0.145560
Nacionality      -6.182e-03   3.060e-03  -2.020 0.043454 *
Mother.s.qualification -1.016e-03   9.815e-04  -1.035 0.300712
Father.s.qualification  7.857e-04   9.709e-04   0.809 0.418474
Mother.s.occupation   5.539e-04   1.196e-03   0.463 0.643239
Father.s.occupation   3.320e-04   1.277e-03   0.260 0.794945
Admission.grade      7.735e-04   1.059e-03   0.730 0.465243
Displaced           -4.370e-02   2.875e-02  -1.520 0.128712
Educational.special.needs -7.446e-02   1.214e-01  -0.613 0.539622
Debtor            -1.548e-01   4.403e-02  -3.517 0.000446 ***
Tuition.fees.up.to.date  4.164e-01   4.505e-02   9.245 < 2e-16 ***
Gender           -4.755e-02   2.784e-02  -1.708 0.087812 .
Scholarship.holder  1.583e-01   3.067e-02   5.160 2.70e-07 ***
Age.at.enrollment  -4.266e-03   2.431e-03  -1.755 0.079439 .
International      3.209e-01   1.317e-01   2.436 0.014918 *
Curricular.units.1st.sem..credited. -3.513e-02   2.102e-02  -1.671 0.094925 .
Curricular.units.1st.sem..enrolled. -3.287e-03   2.444e-02  -0.135 0.893009
Curricular.units.1st.sem..evaluations. -3.977e-03   5.944e-03  -0.669 0.503525
Curricular.units.1st.sem..approved.  8.730e-02   1.465e-02   5.957 2.98e-09 ***
Curricular.units.1st.sem..grade.    -1.162e-02   6.027e-03  -1.929 0.053891 .
Curricular.units.1st.sem..without.evaluations.  2.077e-02   2.320e-02   0.895 0.370851
Curricular.units.2nd.sem..credited.   -6.001e-02   2.250e-02  -2.667 0.007714 **
Curricular.units.2nd.sem..enrolled.   -1.400e-01   2.286e-02  -6.125 1.07e-09 ***
Curricular.units.2nd.sem..evaluations. -1.333e-02   5.793e-03  -2.301 0.021460 *
Curricular.units.2nd.sem..approved.   2.152e-01   1.332e-02  16.154 < 2e-16 ***
Curricular.units.2nd.sem..grade.    -5.533e-03   5.868e-03  -0.943 0.345827
Curricular.units.2nd.sem..without.evaluations.  3.537e-02   2.149e-02   1.646 0.099907 .
Unemployment.rate   -7.241e-03   5.254e-03  -1.378 0.168345
Inflation.rate      -4.492e-03   9.082e-03  -0.495 0.620958
GDP                 5.994e-03   6.200e-03   0.967 0.333783
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

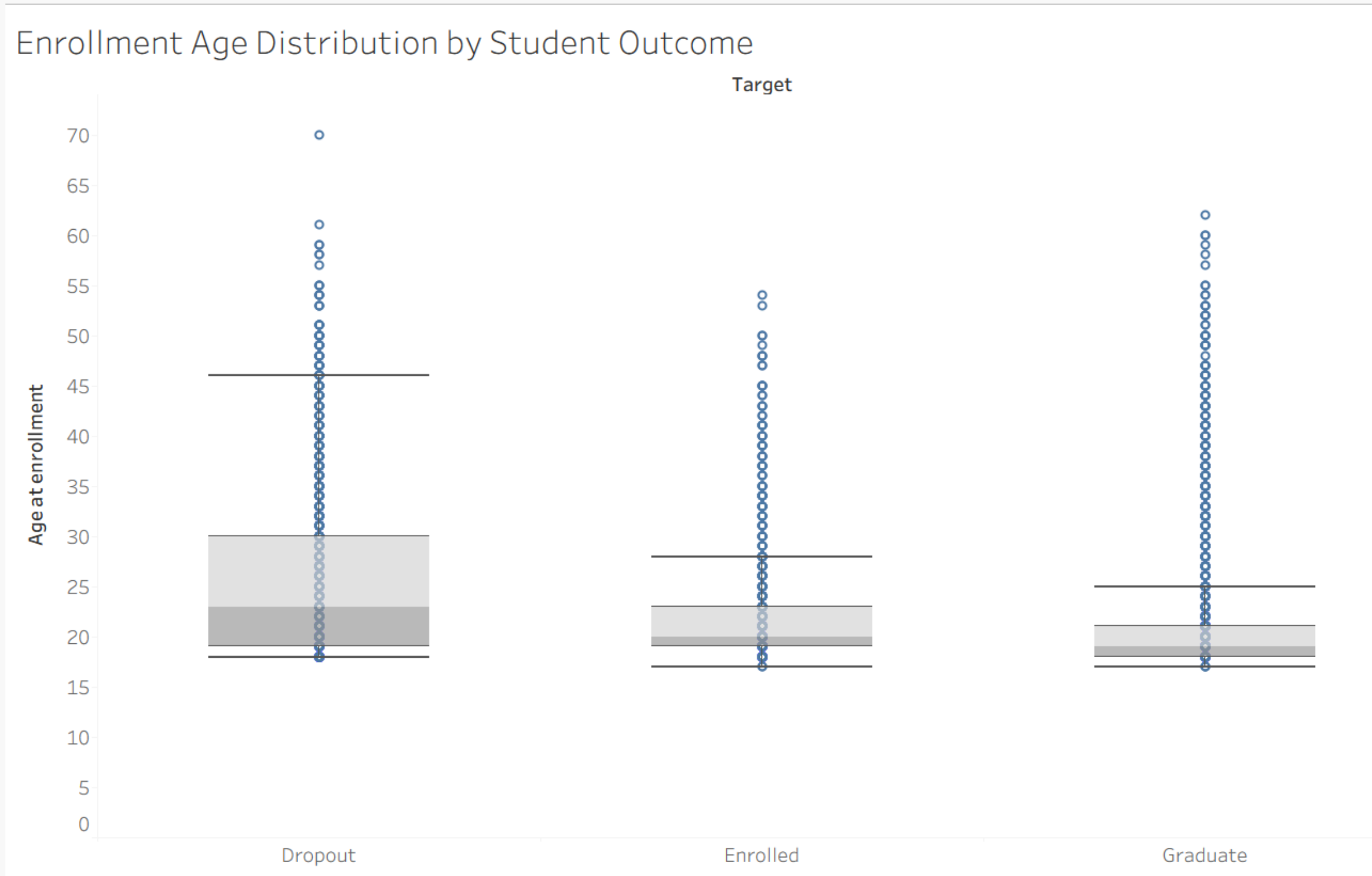
Residual standard error: 0.5746 on 2174 degrees of freedom
Multiple R-squared:  0.5883,    Adjusted R-squared:  0.5815
F-statistic: 86.31 on 36 and 2174 DF,  p-value: < 2.2e-16
```

Significant Variables

- Debtor (does the student already owe money to the school)
- Tuition Fees Up to date (tuition has been paid)
- Scholarship
- Curricular units approved for 1st semester (courses passed in semester 1)
- Curricular units approved for 2nd semester
- Curricular units enrolled in 2nd semester

Multiple R Squared: 0.5883 ← there is a good portion explained, but other models could do better

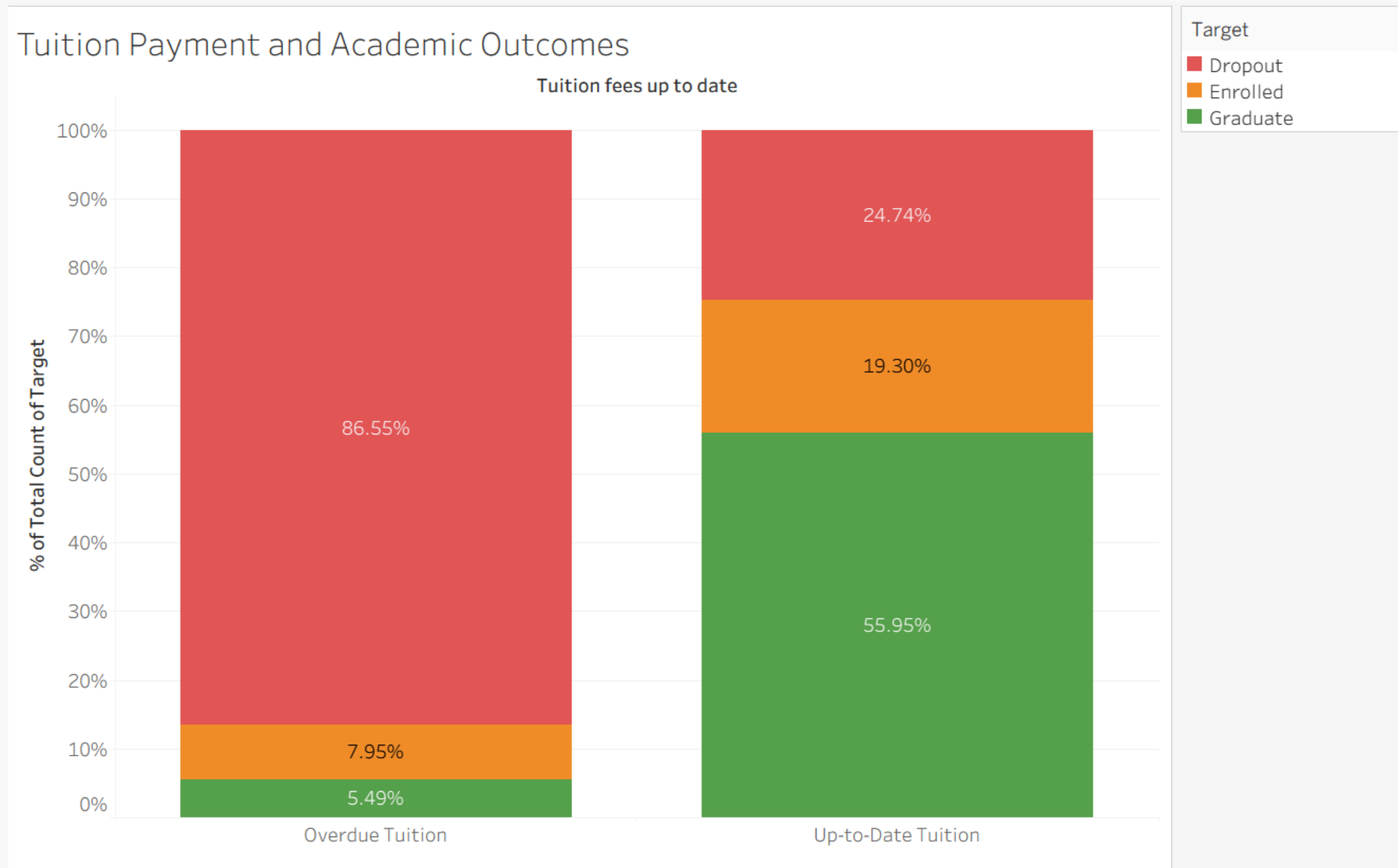
Age and Academic Outcome



Students who dropped out tend to be older at enrollment. Age may be moderately associated with dropout

- Older students may face more pressures such as work, family, etc.
- Age could serve as a supporting variable in predictive modeling

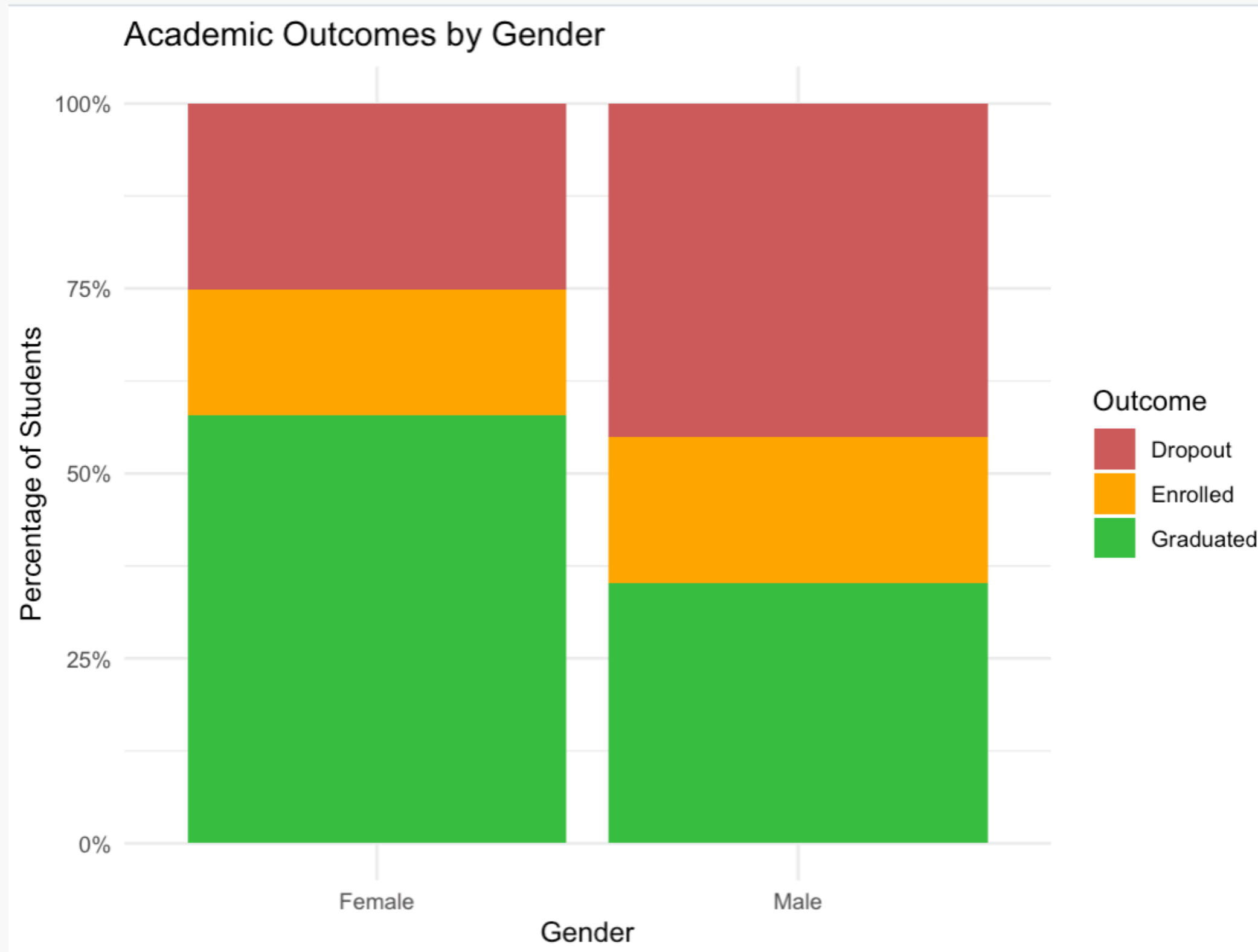
Tuition Payment and Academic Outcome



Students with overdue tuition have a higher dropout rate vs student's with up-to-date tuition are shown to more likely graduate

- Financial stability appears strong with student persitance
- Tuition status may align with graduation likelihood

Gender and Academic Outcome

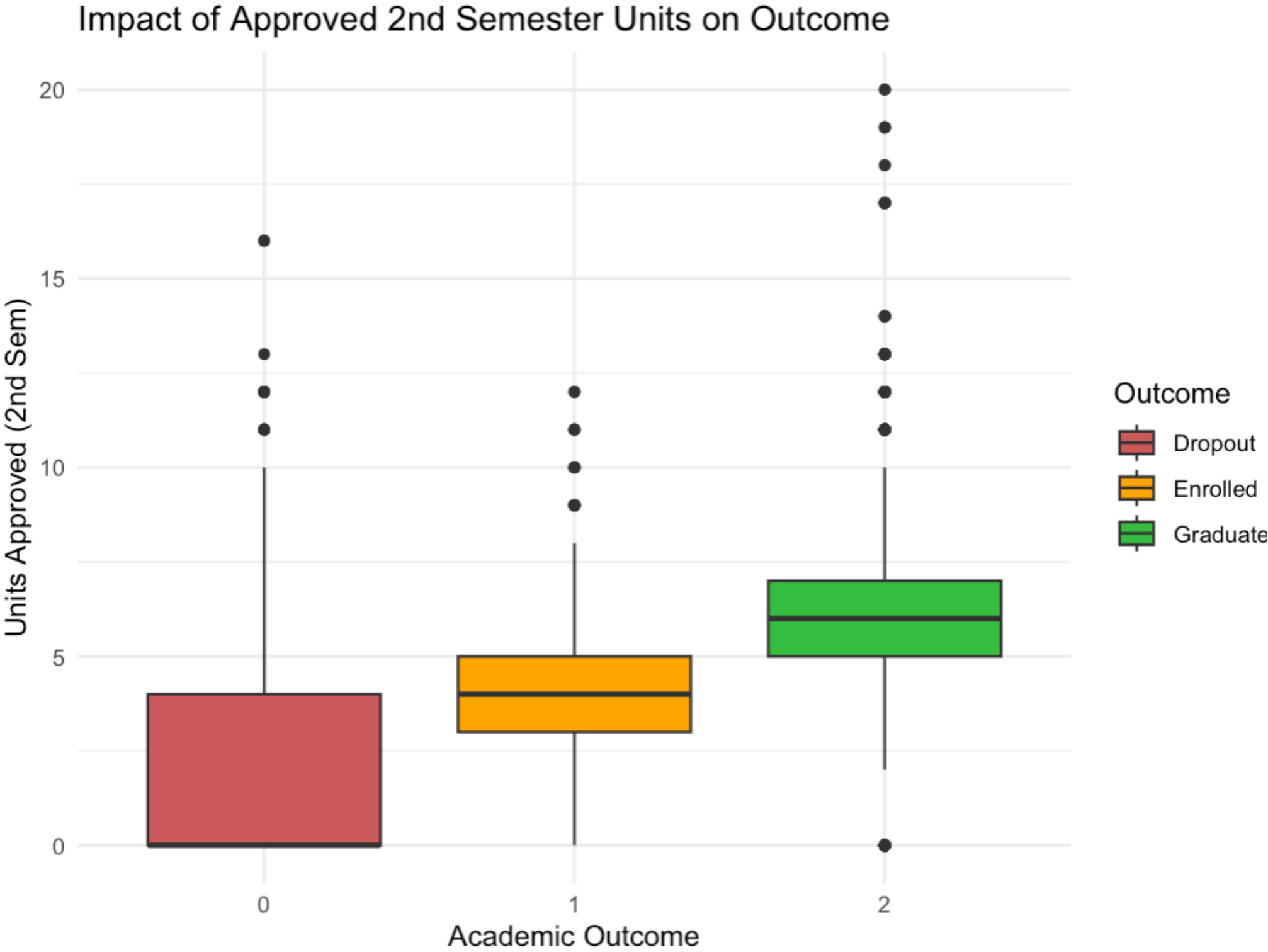


Male students have a significantly higher dropout rate, while female students show a higher graduation rate.

Potential Drivers:

- Lower academic performance (fewer passed units)
- Higher financial pressure (debts/ low scholarship access)

Academic Performance: The Primary Driver of Success



There is a clear positive relationship between curricular units passed and graduation.

Students who graduate have a significantly higher median number of approved units in their second semester compared to those who drop out.

This could be a crucial early-warning indicator.

Summary



Key Findings

- Academic performance is the strongest predictor
- Tuition delinquency strongly signals dropout risk
- Dropout risk is concentrated among older and male students

Next Steps

- Evaluate different predictive models (Logistic, DT, and RF), select best-performing model for outcome prediction
- Identify high-risk students early for targeted intervention



References

Data :

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>.



Thank you

